Lead 1: the n = 3 Tian route. Plane-quartic Jacobians in the class of Res E, and Diem index calculus against rho

How this was produced. A Sonnet lead agent built the tools and ran every measurement under `PROTOCOL.md` (frozen before the runs; Amendments 1–4 dated before the runs they govern). The parent session wrote this report from the agent's findings: the subagent could not write report files. Every number cites a run directory under `results/`.

The parent session re-checked the following:

- both tools pass rustfmt, clippy `-D warnings` and their unit tests (8 and 5);
- the committed protocol matches the file;
- each run's `env.txt` records the protocol version it ran under.

No independent validator has replayed these runs yet.

Answer

- **(a) Jacobian census [exact, sampled].** Theorem 4 (round 2) flags certain prime-order `E/F_{q³}`. For every such class at `q = 7, 11, 13, 17`, the isogeny class of `Res_{F_{q³}/F_q} E` contains the Jacobian of a smooth plane quartic over `F_q`: **26 of 26**.
  - This confirms round 2's untested "the quotient is a Jacobian" step, at the level of isogeny classes. No isogeny was computed.
  - Unflagged prime-order classes are also hit, so a quartic in the class is **not** evidence of a Tian kernel. The flag only changes the hit rate, to about 2×, with caveats below.
- **(b) Diem index calculus on generic quartic Jacobians [measured].** A complete double-large-prime index calculus solved every planted instance up to `q = 8191`, and every answer was verified.
  - With native work priced at 0 it costs 180–340 group additions, against rho's 5,000–300,000.
  - With native work priced at 0.1 and 1 GAE it loses: its dense linear algebra grows like `q^1.4`.
  - **The ranking is unresolved across the §5a band.** This is not a speedup claim, and item (b) is not run on the item (a) classes.

Item (a): census (runs `a-census-run-20261024T2150Z`, `a2-summarize-run-20261024T2228Z`)

Weil polynomial. For `E/F_{q³}` with trace `t3`, Frobenius on `Res E` has characteristic polynomial $P(T) = T^6 - t3 \cdot T^3 + q^3$.

Matching condition. A smooth quartic matches exactly when all three counts agree, and then `#Jac(F_q) = #E`:

- $N_1 = q + 1$;
- $N_2 = q^2 + 1$;
- $N_3 = q^3 + 1 - 3 \cdot t3$.

In that case `c1 = c2 = 0` and `c3 = −t3`.

Irreducibility. `P` is irreducible unless $t3 = s^3 - 3qs$ for some integer $|s| \le 2\sqrt{q}$. Prime `#E` excludes this; the reducible classes occur only for composite order (11 of them at `q = 13`).

Theorem 4, re-implemented independently. The agent's version agrees with round 2's `curves.csv` on class and flag in all 23,182 odd-ℓ rows, with 0 conflicts at the `t3` level.

q	flagged prime-order classes	hit	samples
7	3	3	$3.0 \cdot 10^6$
11	5	5	$3.0 \cdot 10^6$
13	8	8 (fewest hits: 3, at `t3 = 87`)	$3.0 \cdot 10^6$
17	10	10	$2.5 \cdot 10^6$

At `q = 19` all 11 certain-or-ambiguous prime-order classes are hit. They are not counted as positives, because round 2 has no curves there to settle scalar against Jordan.

Negative control (unflagged prime-order classes). These are hit too. Hits by `q`:

q	unflagged classes hit
7	7 / 8
11	10 / 10
13	14 / 14
17	22 / 23
19	26 / 26

The two zeros are `q = 7, t3 = 37` (the Hasse-range edge) and `q = 17, t3 = 127` (local expectation 3.7). Both mean "not found within N", not "absent". Theorem 4 is about Tian kernels, not about Jacobians in the class.

Rate gap. Measured as the mean of hits over local expectation, flagged divided by unflagged; intervals are a bootstrap over `t3`:

q	flagged / unflagged	95% interval
7	2.12	0.90–4.69
11	1.47	1.00–2.12
13	2.07	1.38–3.09

17		1.95	1.26–2.92

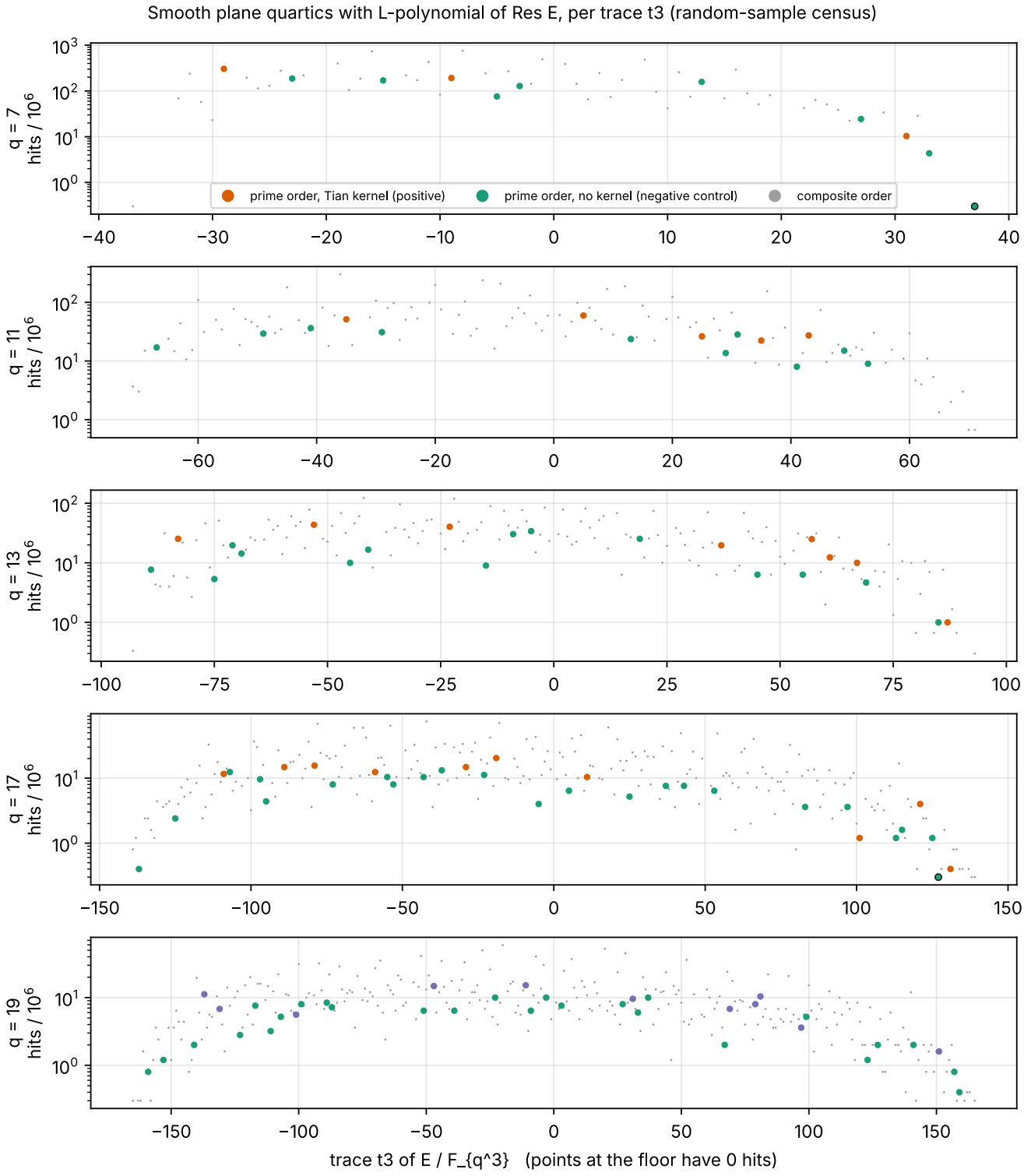
The bootstrap treats the `t3` as independent, and they are not. The local-factor explanation is heuristic and untested.

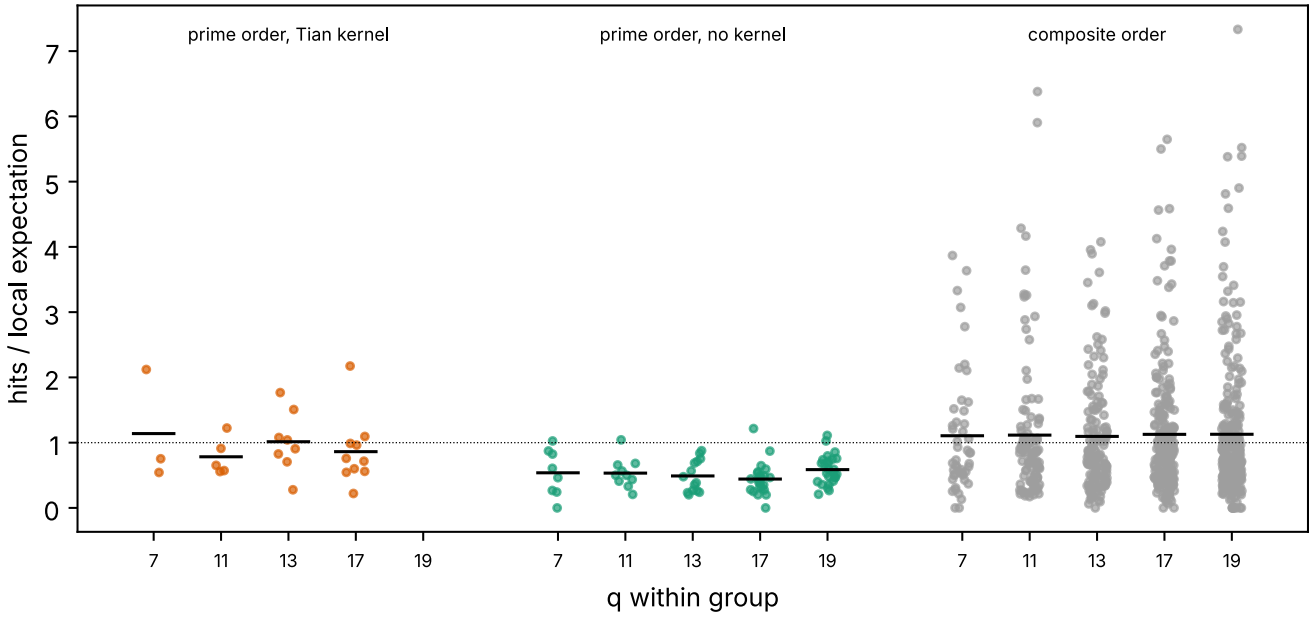
Class masses.

- Smooth quartics: 118,091 at `q = 7` (against q^6 = 117,649), up to 47.1 M at `q = 19`, reproducing q^6 to about 0.5%.
- `P(c1 = c2 = 0 and smooth)`: $1.2 \cdot 10^{-2}$ at `q = 7` down to $3.1 \cdot 10^{-3}$ at `q = 19`.
- Jacobian mass per flagged class: 24, 74, 117 and 272 at `q = 7, 11, 13, 17`, roughly `q^2.7`.

Controls.

- *Positive control.* A constructed quartic `C*` at `q = 7, 11, 13` has brute-force counts `(6, 52, 378)`, `(12, 136, 1323)` and `(11, 189, 2261)`. The fast counters agree, and a fresh search of 10^6 samples finds the same triple 520, 54 and 57 times.
- *Recount.* An independent brute-force recount confirms 35 of 35 stored examples.
- *A/A.* A second seed at `q = 7` gives $\chi^2 = 69.1$ on 55 cells, `z = 1.35`.
- *Smoothness certificate.* Tested against brute-force singular-point search over `F_5`, `F_25` and `F_125`.





Item (b): index calculus against rho (run `b-ic-vs-rho-run-20261024T2246Z`; medians in `b2-median-summary-run-20261024T2258Z`)

Tool. `examples/quartic_ic.rs` provides:

- a group law on the Jacobian of a plane quartic (Mumford-like pairs and two-cubic addition), with negation and doubling;
- orders found by baby-step giant-step;
- double-large-prime index calculus, with the factor base of degree-1 points, relations from lines through two points, a contracted large-prime graph and dense linear algebra mod ℓ;
- Pollard rho, plain and with the negation map.

All arms run in one process and use the same counters.

Instances and verification. There were 12/12/10/10/8/6/4 instances at `q = 127 … 8191`. Every answer was verified: 52/52 for the index calculus, 52/52 for rho and 42/42 for rho with negation. Rho with negation stops at `q = 2039`. One group addition costs about 5,550 M + 229 I.

Mean cost in GAE at `I = 3M`, with native work priced at 0 / 0.1 / 1 GAE (`I = 2·log2 q·M` gives the same table to three digits):

q	index calculus	rho	IC/rho
127	232 / 3,842 / 36,331	5,230 / 5,347 / 6,397	0.044 / 0.72 / 5.7
251	210 / 8,532 / 83,425	4,796 / 5,088 / 7,719	0.044 / 1.68 / 10.8
509	148 / 22,086 / 219,530	9,736 / 11,080 / 23,172	0.015 / 1.99 / 9.5
1021	212 / 66,009 / 658,178	17,918 / 21,086 / 49,596	0.012 / 3.13 / 13.3
2039	177 / 180,870 / 1,807,109	40,744 / 48,529 / 118,592	0.004 / 3.73 / 15.2
4093	202 / 489,622 / 4,894,400	50,927 / 60,902 / 150,681	0.004 / 8.04 / 32.5
8191	338 / 1,411,234 / 14,109,296	302,417 / 363,186 / 910,108	0.001 / 3.89 / 15.5

At `q = 8191` one rho instance of $1.1 \cdot 10^6$ steps inflates the mean. With medians the IC/rho ratios there are 0.007 / 27 / 109.

Fitted slopes in `q` (finite range, 7 values of `q`, intervals 95%):

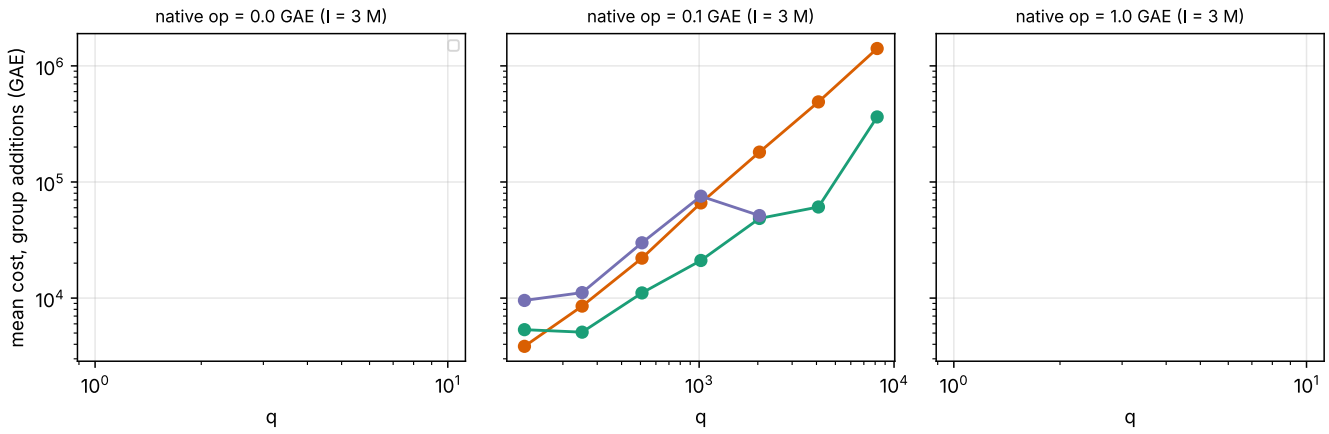
quantity	slope	interval	reading
IC cost, native at 0	0.06	−0.02–0.16	about 120–180 target additions, independent of `q`
IC cost, native at 0.1 / 1	1.43 / 1.45	1.41–1.47	dense linear algebra, about `27·q^1.5` multiply-adds
rho cost, native at 0	0.94	0.48–1.13	pulled down by a set-up of about 4,000 additions
rho steps	1.42	—	theory: 1.5
IC relation stage only	0.69	0.68–0.70	a stage diagnostic, not a speedup

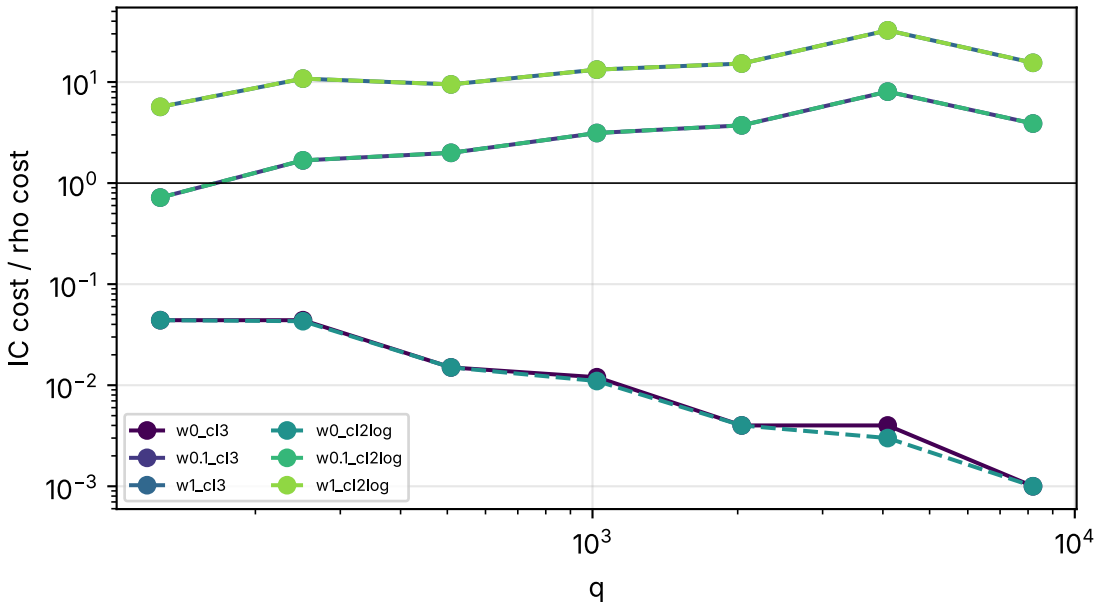
Ranking (AGENTS.md §5a). The index calculus wins with native work priced at 0 and loses at 0.1 and 1, for every `q ≥ 251`, so every pair is **unresolved**.

Predictions.

- P-b1: holds.
- P-b2 (relation-stage slope 1.0–1.3): not supported. The stage measures 0.69 in GAE units and about 0.9 in raw M.

- P-b3 (no resolved ranking): holds.
- P-b4: mixed. Rho with negation costs 1.7–4× plain rho at `q ≤ 1021`, and about the same at `q = 2039`.





Distance to rho

Item (a) has no cost rows. For item (b), IC/rho is:

- 0.001–0.044 with native work at 0;
- 0.7–8.0 at 0.1 GAE;
- 5.7–32.5 at 1 GAE.

No row puts the index calculus below rho at every weight. The relation stage alone sits below rho at every weight, but that is a stage diagnostic.

What is not established

- **No end-to-end attack.** No Tian isogeny `Res E → Jac(C)` was computed, so no attack on `E/F_{q³}` was run. Only the existence of a quartic in the isogeny class is shown, and only for `q ≤ 19`.
- **Item (b) uses generic quartics, not the item (a) classes.**
- **No cost exponent.** Seven primes, a 65× range in `q`, a rho set-up constant and a varying cofactor allow finite-range slopes only.
- **Dense linear algebra.** A sparse solver (about `B²·d` instead of `B³`) was not measured, and is the likeliest change to the 0.1 and 1 GAE cells.
- **`B = 3√q` is not optimised.**
- **The rate-gap explanation is untested.**
- **Instance filter.** A planted pair is accepted only if small multiples are computable (Amendment 4). How many pairs were rejected was not recorded.
- **Not yet admitted.** No independent replay has been run, so this study is not in the results table.

Records kept

- `b-ic-vs-rho-run-20261024T2228Z`: the first run. Its rho-with-negation rows have no two-cycle escape.
- `b-ic-vs-rho-run-20261024T2234Z`: aborted at `q = 509` on an unusable `Q`.

Both are superseded by the `…T2246Z` run.

Next improvements

1. Sparse linear algebra mod ℓ (block Wiedemann or Lanczos), counted.
2. A cheaper target phase, since the target additions are now the whole cost with native work at 0.
3. Connect (a) and (b): compute `E/F_{q³} → Jac(C)` at `q ≤ 19` with a matched rho in `E`, so the Tian route is priced end to end.
4. Item (a) at larger `q`, with a rank test of the rate gap.

Reproduce
`run_item_a.sh`, `run_item_b.sh`; `sh build.sh` builds this PDF.